MATHS MANTHAN
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**Subject:** Mathematics | **Periodic Test Paper 12 (Cumulative)** | **Time:** 60 Minutes
**Class:** Class X (CBSE) | **Chapter 5: Arithmetic Progressions** | **MM: 30**

**Student Name:** _____ **Roll No.:** _____ **Date:** _____

**General Instructions:**

- This question paper consists of **12 questions** divided into **Five Sections**: A, B, C, D, and E.
- **Section A** comprises **5 questions** of **1 mark** each (Q1–Q4 MCQs, Q5 Assertion-Reason).
- **Section B** comprises **2 questions** (Q6–Q7) of **2 marks** each.
- **Section C** comprises **2 questions** (Q8–Q9) of **3 marks** each.
- **Section D** comprises **2 questions** (Q10–Q11) of **5 marks** each.
- **Section E** comprises **1 Applied Case Study** (Q12) of **5 marks** with internal choice in sub-part (iii).
- All questions are compulsory. Use of calculators is strictly prohibited.

## SECTION A: Multiple Choice & Assertion-Reason Questions (5 × 1 = 5 Marks)

**Q1.** The $30^{\text{th}}$ term of the AP $10, 7, 4, \ldots$ is: [1]

(A) 97 (B) 77 (C) $-77$ (D) $-87$

**Q2.** The middle term of the finite AP $213, 205, 197, \ldots, 37$ is: [1]

(A) 125 (B) 117 (C) 121 (D) 129

**Q3.** Find the sum of all three-digit natural numbers which are divisible by 7: [1]

(A) 70336 (B) 70350 (C) 70314 (D) 70320

**Q4.** The sum of the first 20 odd natural numbers is: [1]

(A) 100 (B) 200 (C) 400 (D) 420

**Q5.** **Assertion (A):** In an AP with first term $a$ and common difference $d$, the $n^{\text{th}}$ term from the end is given by $l - (n-1)d$.
**Reason (R):** Counting backwards from the last term $l$, the common difference becomes $-d$. [1]

**(A)** Both (A) and (R) are true, and (R) is correct explanation.
**(B)** Both (A) and (R) are true, but (R) is not correct explanation.
**(C)** Assertion (A) is true, but Reason (R) is false.
**(D)** Assertion (A) is false, but Reason (R) is true.

## SECTION B: Very Short Answer Questions (2 × 2 = 4 Marks)

**Q6.** Find the $4^{\text{th}}$ term of the AP whose sum of first $n$ terms is given by $S_n = 2n^2 + 3n$. [2]

**Q7.** Find the sum of the first 20 terms of an AP in which the common difference $d = 7$ and the $20^{\text{th}}$ term is 143. [2]

## SECTION C: Short Answer Questions (2 × 3 = 6 Marks)

**Q8.** The $8^{\text{th}}$ term of an AP is equal to three times its $2^{\text{nd}}$ term, and its $12^{\text{th}}$ term exceeds twice its $5^{\text{th}}$ term by 1. Find the AP. [3]

**Q9.** Find the sum of all natural numbers between 250 and 1000 which are exactly divisible by 3. [3]

## SECTION D: Long Answer / HOTS Questions (2 × 5 = 10 Marks)

**Q10.** If $S_n$ denotes the sum of first $n$ terms of an AP, prove that:

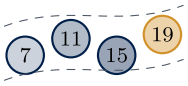
$$S_{12} = 3(S_8 - S_4)$$

[5]

**Q11.** The sum of the first 7 terms of an AP is 49 and that of the first 17 terms is 289. Find the sum of its first $n$ terms, and hence calculate $S_{25}$. [5]

## SECTION E: Applied Case Study (1 × 5 = 5 Marks)

### Q12. CASE STUDY: Decorative Stepping Stones in Landscaped Walkway




A landscape architecture firm is contracted to lay out a paved promenade across a botanical garden.
The walkway incorporates circular numbered stepping stones engraved with sequential serial numbers that follow an arithmetic progression. The sequence of numbers on consecutive stones begins with $7, 11, 15, 19, \ldots$.
The pathway features 40 stones in total.

Based on the above architectural installation, answer the following questions:

(i) What number is engraved on the $25^{\text{th}}$ stepping stone? [1]

(ii) Which stepping stone in the promenade is engraved with the number 115? [1]

(iii) Calculate the sum of all serial numbers engraved on the 40 stepping stones ($S_{40}$). [3]

**OR**

(iv) If a shorter segment consisting of only 31 stepping stones is considered, find the number engraved on the central (middlemost) stone. [3]

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